Quiz 5 — Instructor Key

Friday, October 16, 2026 10 points

1. (3 points)

(a) $g'(x) = 12x^2 - 5$,

(b) $r(x) = 6x^{-2}$, $r'(x) = -12x^{-3} = -\dfrac{12}{x^3}$,

(c) $s(x) = 3x^{1/2} + 2x$, $s'(x) = \dfrac{3}{2}x^{-1/2} + 2 = \dfrac{3}{2\sqrt{x}} + 2$.

Accept equivalent forms. Give partial credit for a correct power rewrite or correct individual terms when the final derivative contains an error.

2. (2 points) $f'(x) = 2x - 3$, so $f'(1) = -1$. The tangent line through $(1, 0)$ with slope -1 is $y - 0 = -(x - 1)$, or $y = 1 - x$.

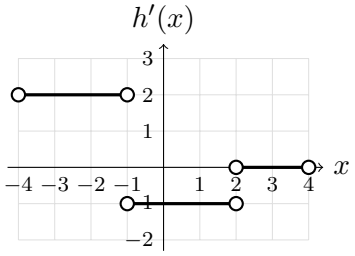
3. (2 points) (a) At a production level of 100 notebooks, total cost is increasing at a rate of \$12 per notebook. The next notebook therefore costs approximately \$12 to produce.

(b) $L(q) = 2500 + 12(q - 100)$ dollars, so $C(103) \approx L(103) = 2500 + 12(3) = \$2{,}536$.

4. (3 points) The slopes of the three line segments are 2, -1, and 0, respectively. Thus, for $-4 < x < 4$,

$$h'(x) = \begin{cases} 2, & -4 < x < -1, \\ -1, & -1 < x < 2, \\ 0, & 2 < x < 4. \end{cases}$$

At $x = -1$ and $x = 2$, the left and right slopes disagree, so h' does not exist. No value is assigned to h' at either corner.



Grading: carry a minor arithmetic error forward without penalizing it again when the later reasoning is correct.